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Achieve Consistently Higher ROP in Factory Wells Performance by Utilizing Push-The-Bit Rotary Steerable System with Automation Drilling Platform, A Case Study from the First Run in Onshore South-Sumatra Basin, Indonesia

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Abstract

Onshore field of South Sumatra basin are known as one of the largest oil petroleum systems in Indonesia. Its production rates are considerably high and healthy as there are multiple target layers of sandstone reservoir with sustainable pressure (Galih, et al.2023). This natural ease had levelled up the confidence to do extraction with factory wells design, means applying similarity of drilling operations (such as trajectories profile and BHA configuration) for several areas in the field. The significant amount of time in preparation was reduced with this method. The next level of time saving comes from maximizing the drilling operations in each well. The case study in this paper will elaborate the initial implementation of drilling technology known as intelligent rotary-steerable-system (RSS) push-the-bit to improve each well delivery time.

Two types of wells, which represent the most common design of trajectory in this subject field, were selected to implement this novel technology. On a J-Type well with target inclination of ~ 22 deg, RSS capabilities are reviewed in the 12.25in hole section by gauging the performance in maintaining tangent interval of ~ 770 m. In a S-Type well with maximum inclination of ~ 18 deg inclination, RSS was evaluated in the 8.5in hole section on holding a verticality until the end of well for ~ 785 m. Both cases were having a comprehensive logging-while-drilling (LWD) suite consisting of gamma ray, resistivity, density, neutron porosity, pressure-while-drilling, and vibration mitigation tool. In addition, this technology is equipped with an automation platform which enables autosteer services to achieve precise steering with a reduced tortuosity borehole, while simultaneously increasing the rate-of-penetration (ROP). Offset wells analysis of each field had helped to form optimized configuration for the RSS. Identification of formation hardness, lithology lamination, loss, washout and tight hole hazard zones, reactive shales, and hard stringers were all considered in the optimization of RSS hydraulics design.

Overcoming respective challenges, Target Well-1 has successfully achieved a significant ROP of ~ 65 m/hr with precise placement of 0.15 m center-to-center to plan and average dog-leg-severity (DLS) at ~ 0.5 deg/30m in the 12.25in hole. Laminated formation with massive change in strength properties, was a major challenge in this Field-G that may cause steering drift. Performance of automation platform plays

a prominent role in achieving quick and adaptive control. Intelligent RSS ability of applying automated trajectory control, coupled with active machine learning algorithm had managed to secure precise placement in this well. In Field-B, Target Well-2 successfully achieved up to ~80 m/hr with 1.55 m center-to-center to plan and average of DLS 0.2 deg/30m in 8.5in hole. Excessive torsional vibration is a repetitive obstacle in this Field-B environment and has reduced the BHA outcome in several wells. The stability and consistency of RSS performance throughout this run had enabled a good quality borehole. intelligent RSS and the autonomous drilling platform predicting the optimum commands to create a smooth wellbore.

A new benchmark was set in this entire factory wells project. These first runs in each field have proven the capabilities to push time and cost savings by applying novel technology in drilling operations. The case studies show the proven RSS can produce benefits of up to 3 times higher delivery time compared to the surrounding offset wells that utilize conventional mud displacement motor. Future applications towards the entire campaign may significantly be a game-changer in onshore drilling operations.

Introduction

Indonesia's Major Field: South Sumatra Basin

South Sumatra province is known to hold the largest onshore drilling operations in Indonesia, occupying mainly two main area operations called Prabumulih (east part) and Pendopo (west part) ([Aprilian, et al. 2003](#)). Activities of oil and gas extraction started in 1920s and reached the peak of oil production in 1954 up to ~30,000 BOPD. To date, more than ~800 production wells and ~280 injector wells have been drilled across ~7 major fields and ~60 structures, displaying declining rate down to ~24,153 BOPD ([Galih, et al. 2023](#)). With the current level of maturity, where massively coming from high water cut in production phase, cost savings are expected to be conducted from all aspects of operations. Injector wells, artificial liftings, targeting tight oil reserves, are all to name a few of foremost directions taken by the operator in subsurface level which helped to increase each well production to ~200 BOPD ([Fansuri, et al. 2021](#)). On the other side lies drilling operations, the initiative to reach a fast sequence between each well operations had pushed the entire field to have a scheme of factory wells design and execution. The novel method to start saving costs in this segment is by implementing effective well planning and faster drilling performance. This paper will elaborate the utilization of the emerging technology in directional drilling known as the RSS push-the-bit technology.

For additional considerations, the environment of South Sumatra basin consist of several drilling hazards driven by the geological structure, some as expected are: bit balling and coal drilling in upper interval of 17.5" hole section; sloughing shale and pack-off in intermediate 12.25" hole section; coal drilling, differential sticking and lost circulation with high potential of pumping LCM (lost-circulation-material) through BHA in reservoir 8.5" hole section.

RSS Push-the-Bit for Faster Performance

With all the latest technology emerging in the field of drilling, fit-for-purpose selection has become very essential. RSS as a main driver in drilling BHA had gone a massive development in past decades, specifically revolving around the basic mechanism of push-the-bit. Robust piston inside the body of RSS push-the-bit had enabled a forceful pad (located as three pieces in circumference) to push against the formation, enabling powerful yet smooth mechanical movement on steering an entire BHA. Faster steering is promptly achievable with high mechanical specifications ([Fig.1](#)), several to mention are the endurance of up to 400 revolution-per-minute (RPM); 65,000 lbf of WOB; 18,500 ft-lbf of torque; and up to maximum 18 deg / 100 ft of DLS (dog-leg-severity). Proprietary self-diagnosis and self-prognosis electronics have also allowed for accurate steering and improved reliability. Most of all, automation-enabled is what makes this technology levelled at an intelligent state, delivering predictable results supplementing effective directional control along with real-time vibration management ([Adiandra, et al. 2025](#)).

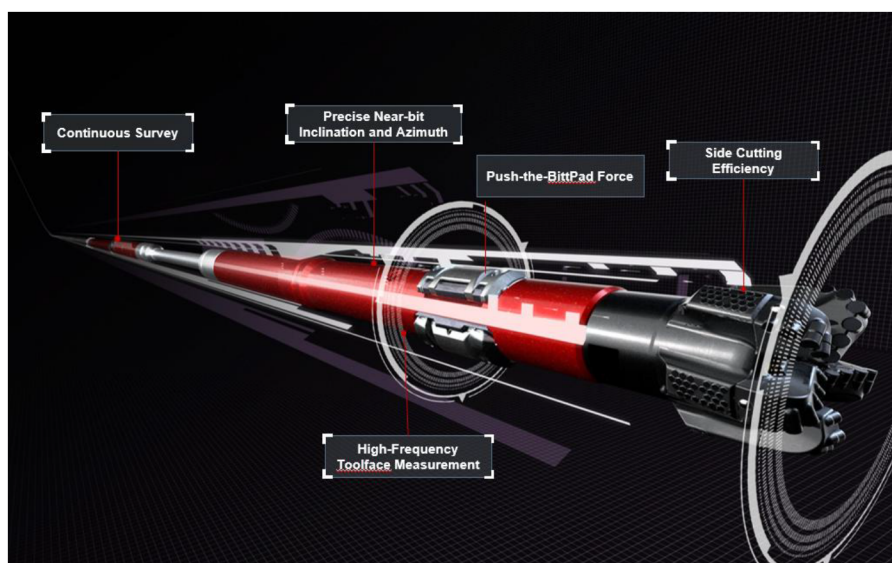


Figure 1—Intelligent Rotary Steerable System Push-the-Bit mechanism component and feature

Automation Drilling

Real-time decisions constantly impact well placement and wellbore quality and an incorrect decision from miscalculations, lack of experience, or outside influences can have costly results such as inconsistent and poor well delivery, or HSE incidents. The Autonomous Drilling Platform offers an integrated autonomous drilling solution that drills smart wells consistently, with accurate well-positioning, and superior drilling performance. The platform orchestrates real-time steering controls, collision avoidance, visualization, and autonomously makes decisions to mitigate drilling dysfunctions and help maximize penetration rates. With the platform, a digital transformation that helps reduce operational risk and uncertainty to deliver wells reliably, repeatably, and consistently. Automation in drilling activities comes as digital twin of the BHA – project well path, to make smart decisions, achieve predictable and consistent results (Fig.2).



Figure 2—Intelligent RSS Push-the-Bit automated-enabled technology

Consistent Well Delivery, the platform provides an autonomous drilling system that uses a digital twin and machine learning to steer without human intervention resulting in consistent well delivery. The platform helps significantly reduce operational and HSE risks. Environmental sustainability and carbon

emission targets can be met with smaller well-site footprints and reduced well delivery times. *Accurate and Repeatable Well Positioning*, through the closed-loop control system of the platform quickly reacts as drilling conditions change downhole using sophisticated algorithms combined with high frequency downhole data. By autonomously adjusting drilling parameters, the well path, or both, the platform ensures the wellbore stays within the desired targets. Continual updates of the downhole environment allow the platform to deliver repeatable and accurate well positioning. *Superior drilling performance*. platform integrates autonomous drilling solutions that help enable superior drilling performance on every well. The steering solution autonomously steers the well trajectory, the drilling dynamics solution mitigates vibration while drilling, and the collision avoidance solution provides monitoring and alerts to facilitate safe drilling in environments with a high risk of collision.

Case Study: Intelligent RSS Push-the-Bit with Automation Platform for Factory Wells

Pre-Job Planning: Optimization in Design-of-Services (DOS)

There are two Target Wells to be elaborated in this paper; both are samples of the common factory wells trajectory design in this subject field, J-Type and S-Type Wells with challenge of maintaining tangent interval in low inclination. The main objective is to increase rate of penetration (ROP) and ensure precise steering and high-quality borehole. Strategic approach had been enhanced across several key areas, from pre-job planning of well trajectory and BHA engineering simulation; up to tight monitoring of real-time execution. These include the combination of advanced directional drilling tools, optimization of drilling parameters, and proactive management of borehole conditions.

A design-of-service (DOS) document is developed as the primary guideline of the planning and execution of directional drilling operations. It serves as a comprehensive guideline that governs all aspects of the drilling process, from the planning stage, engineering analysis and defines the customized service configuration to ensure that all stakeholders are aligned on the objectives, methods, and strategies for efficiently drilling a high-quality well.

Table 1—Design of Services for Conventional Positive Displacement Mud Motor vs Rotary Steerable System (RSS) Push-the-Bit

Design of Services Aspect	Conventional Mud Motor	Rotary Steerable System Push-the-Bit
Trajectory Design	Input: - DLS range PDM (dependent to bend settings). - Correction interval based on record BHA tendency	Input: - DLS range RSS Push-the-bit (up to 14 deg for Tool 8" OD; and up to 21 deg for Tool 6.75" OD). - RSS Automated Trajectory control ability.
Anti-Collision	Strict adherence against safety factors towards offset well	Strict adherence against safety factors towards offset well
BHA Design and Modelling	- Bend motor selection vs Motor output	- RSS hydraulic setup for building or dropping capability - Flex collar utilization to enhance building or dropping capability
Torque and Drag Analysis	- More required of washing and reaming	- Less required of washing and reaming
Hydraulics Analysis	- Motor Differential pressure will be added to estimate total pressure	- Less difference between on bottom and off bottom pressure

Target Well (1) RSS Push-the Bit Performance for 12.25in HS

Realtime Execution: Precision in Tangent Section with RSS Technology

Plan vs Actual. Well-1 was planned to be drilled using a push-the-bit RSS in the 12.25in hole section. This 12.25in hole section is a tangent section with an inclination of 22.56 degrees and a planned DLS of 1°/30 meters, covering an approximate length of 770 meters. Due to directional and well placement requirements, the center-to-center (CtC) distance from the planned trajectory must not exceed 5 meters throughout this section.

The use of push-the-bit RSS in this context enables precise trajectory control while maintaining continuous rotation of the drill string, which is essential for achieving the required wellbore placement accuracy and minimizing tortuosity. The specified trajectory and tight center-to-center (CtC) constraint emphasize the need for high precision directional control, a capability for which RSS technology is particularly well suited.

The implementation of the push-the-bit RSS in the 12.25in hole section of Well-1 resulted in highly accurate directional control, effectively meeting the planned trajectory requirements. The system consistently maintained a center-to-center (CtC) deviation of less than 1 meter from the planned path and operated within a maximum observed dogleg severity (DLS) of 1.7°/30 meters. These results demonstrate the technical suitability and operational reliability of push-the-bit RSS technology for delivering high precision wellbore placement in tangent sections with strict directional requirements.

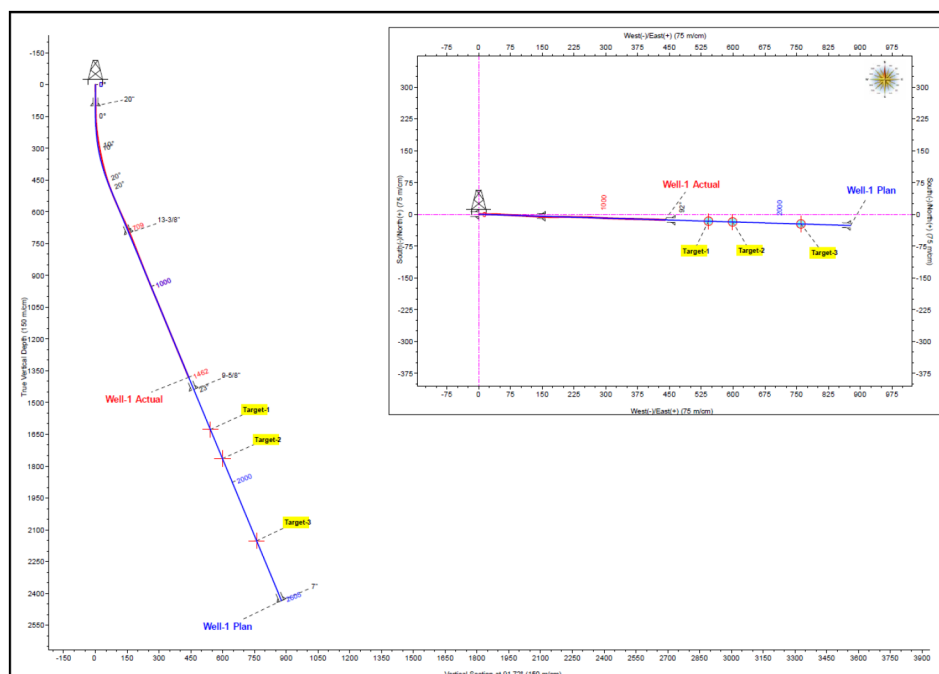


Figure 3—Plan vs Actual Trajectory

Wells Drilled with Conventional Mud Motor vs Rotary Steerable System (RSS). The drilling operation in the 12.25in hole section successfully achieved 755 meters drilled interval, demonstrated that the push-the-bit RSS significantly outperforms the Conventional Mud Motor (PDM) in terms of ROP, overall drilling efficiency, and tool performance. The RSS's ability to deliver continuous rotation while maintaining directional control combined with higher RPM and lower WOB resulted in improved hole quality and reduced non-productive time (NPT). While the Conventional Positive Displacement Motor (PDM) remains

a viable and lower-cost alternative in certain applications, the operational advantages of RSS make it the preferred choice in long tangent sections requiring consistent trajectory control and optimized ROP.

Fig.4 presents a comparative evaluation of key drilling parameters in the 12.25in hole section, comparing the performance of the push-the-bit RSS used in Well-1 with the Conventional PDM used in Offset Well-1.

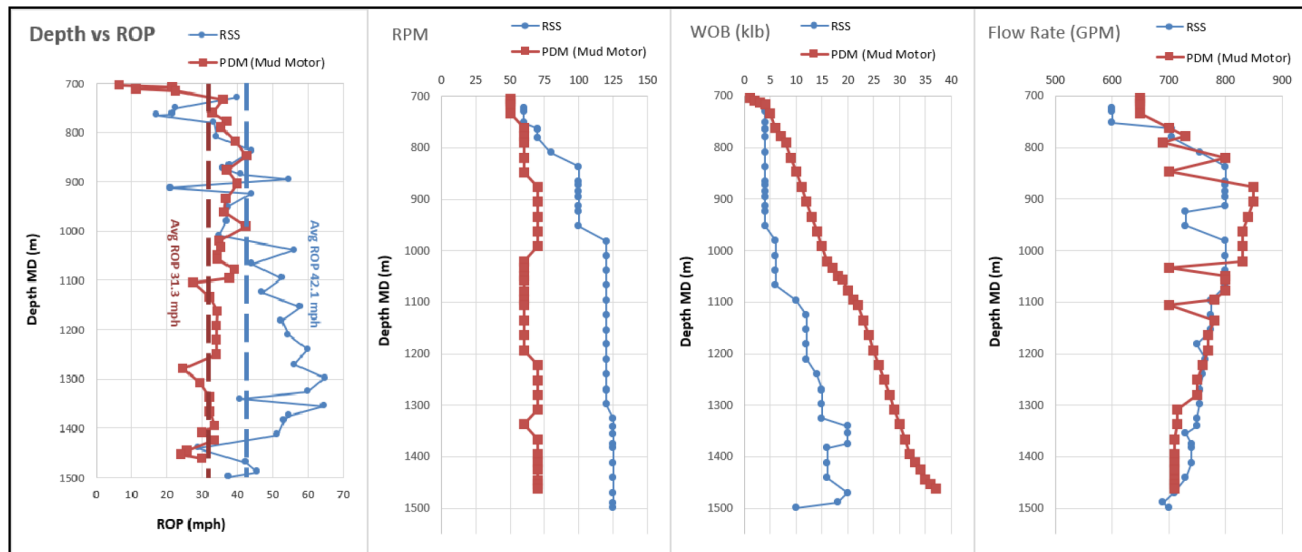


Figure 4—Performance Comparison (ROP, RPM, WOB, Flow Rate) between Target Well-1 and Offset Well-1

The first track (Depth vs. ROP) shows that the push-the-bit RSS consistently achieved higher Rates of Penetration (ROP), especially from approximately 900–1550 m depth, where ROP frequently exceeded 60 m/hr. In contrast, the PDM delivered lower and more variable ROP, typically below 45 m/hr over the 700–1150 m interval. The second track (RPM vs. Depth) demonstrates that the push-the-bit RSS operated with higher surface-driven RPM, throughout its interval, which contributed to better bit rotation, reduced stick-slip tendencies, and improved hole cleaning. In contrast, RPM with the PDM was significantly lower, constrained by the tool's internal rotor-stator mechanics and its dependence on flow rate. The third track (WOB vs. Depth) indicates that the PDM required higher WOB, ranging from 20 to 35 klb, to maintain drilling progress. On the other hand, the push-the-bit RSS maintained effective drilling performance with WOB typically below 15 klb, reflecting better energy transfer to the bit and reduced mechanical stress on the drilling assembly. The fourth track (Flow Rate vs. Depth) shows that the push-the-bit RSS operated at higher flow rates, ranging from 750 and 900 GPM, which supported both tool functionality and improved cuttings transport. The PDM, limited by tool-specific pressure drops and operational constraints, maintained lower flow rates, typically in the 650 to 800 GPM range.

Comparison Against Several Offset Wells. By utilizing ROP data from wells that use both push-the-bit RSS and Conventional Mud Motors (PDM), it is possible to accurately calculate and compare drilling time between the two technologies.

The observation from Fig.5 that illustrates the Time vs. Depth profiles for Well-1 and several offset wells in the 12.25" hole section shows that Well-1, which was drilled using a push-the-bit RSS, achieved significantly faster drilling times compared to the offset wells, which were drilled using Conventional PDM. This clearly indicates the performance advantages of RSS technology. Specifically, Well-1 completed the drilling in 18 hours, achieving the highest average ROP among all offset wells in the subject field at 42.1 m/hr. In contrast, the offset wells required 6 to 17 additional hours to drill similar intervals, with average ROPs ranging from 20.4 to 30.8m/hr. These results highlight the operational efficiency and time saving benefits of utilizing push-the-bit RSS in extended tangent sections.

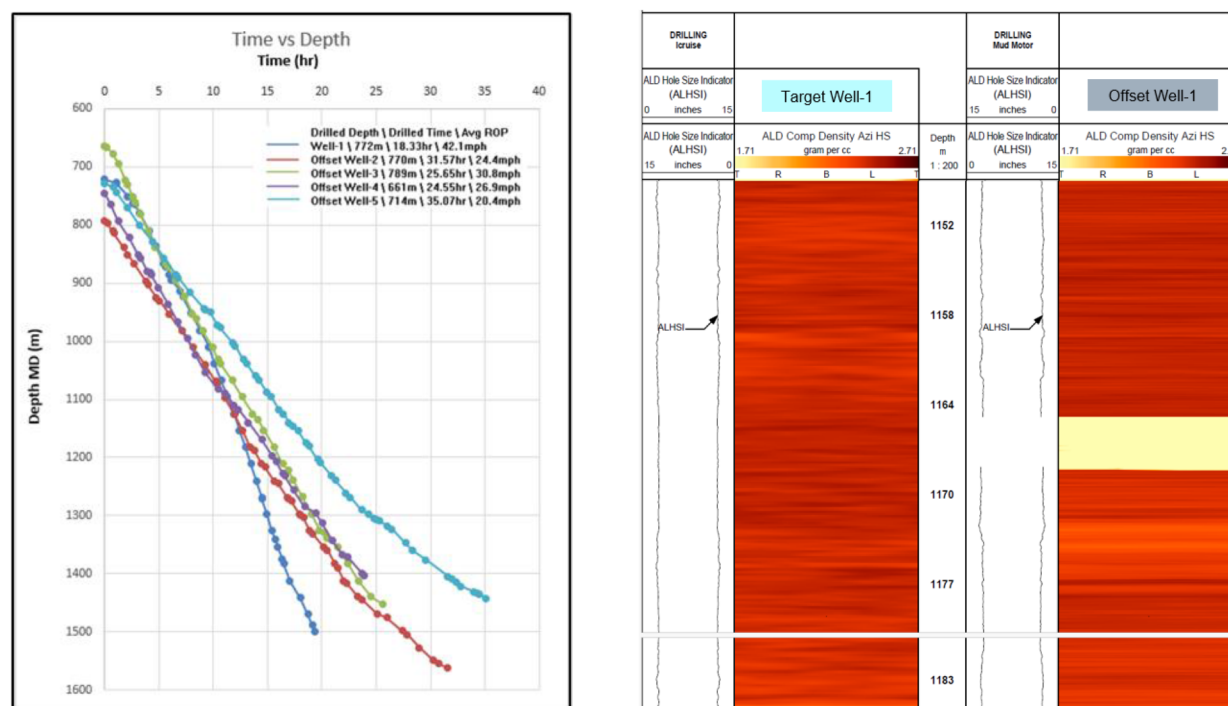


Figure 5—(Left) Performance Comparison (Time vs Depth) against several offset wells; (Right) Caliper and Density Image displays Borehole Quality

Borehole Quality. Borehole quality plays a crucial role throughout both the drilling phase and post drilling operations. During the drilling process, maintaining good borehole quality helps minimize risks such as stuck BHA and Lost in Hole (LIH) incidents, leading to costly delays and potential safety hazards. (Fig 5.) Shows that with RSS push-the-bit, tortuosity gauge by caliper and density image are able to be steadily maintained; while using Mud Motor, not only washout were frequently to happen, but also density image measurement seems to be interrupted due to the sliding activities being conveyed. Once drilling is complete, borehole quality continues to be important, especially during the casing running process. A well-maintained borehole allows casing to be run to total depth without excessive drag, resistance, or risk of damage to the casing string. High quality boreholes contribute to overall integrity, reduce non-productive time (NPT), and enhance operational efficiency.

Target Well (2) Push-the Bit Performance for 8.5in HS

Realtime Execution: Precision in Vertical Drilling with RSS Technology

Plan vs Actual. The execution of vertical drilling in the 8.5" hole section of a S-Type well—with a maximum inclination approaching 18°—was successfully accomplished using a RSS configured for vertical hold mode. Drilled interval of 785 meters, from 1579 m to 2364 m MD.

The drilling assembly with LWD tools include gamma ray, resistivity, density, neutron porosity, pressure-while-drilling, and vibration mitigation sensors. These tools supported real-time decisions and offered full formation insight for proactive control of borehole conditions, steering mode with Vertical Trajectory Control.

The result was a center-to-center deviation from plan of just 1.55 meters, indicating exceptional vertical delivery across a complex trajectory and geological hazard. DLS was also effectively controlled, with an average of 0.20°/30m, suggesting a smooth and minimally tortuous wellbore path. The rate of penetration (ROP) remained stable, averaging 47.2 m/h, and peaking at 79 m/h during high-efficiency intervals, with

rotary speeds (70–170 RPM) and gradual weight on bit (25–32 klb). Flow rates around 550–600 GPM range to help optimize hole cleaning.

The successful delivery of this vertical hole section shows the capabilities of RSS push-the-bit systems when applied to strategic planning and proactive real-time execution. With only 1.55 m deviation from plan and a stable ROP profile over a 785-meter MD vertical section (Fig. 6).

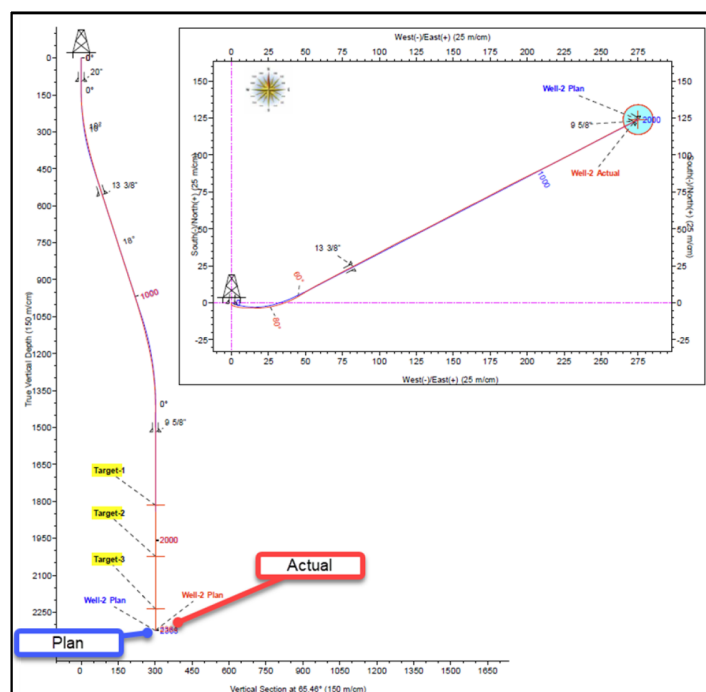


Figure 6—Precision steering achieved by implementing adjustments in planning stage and result of Benuang Project Directional Drilling Plan vs. Actual (design-of-services).

Directional Drilling Performance Evaluation: RSS Push-the-bit vs. PDM (Mud Motor). The performance characteristics of the push-the-bit RSS and a traditional PDM (Mud Motor) were analyzed based on actual drilling parameters, including ROP, RPM, WOB, and Flow Rate (FR). These parameters were plotted against depth to evaluate operational efficiency and consistency.

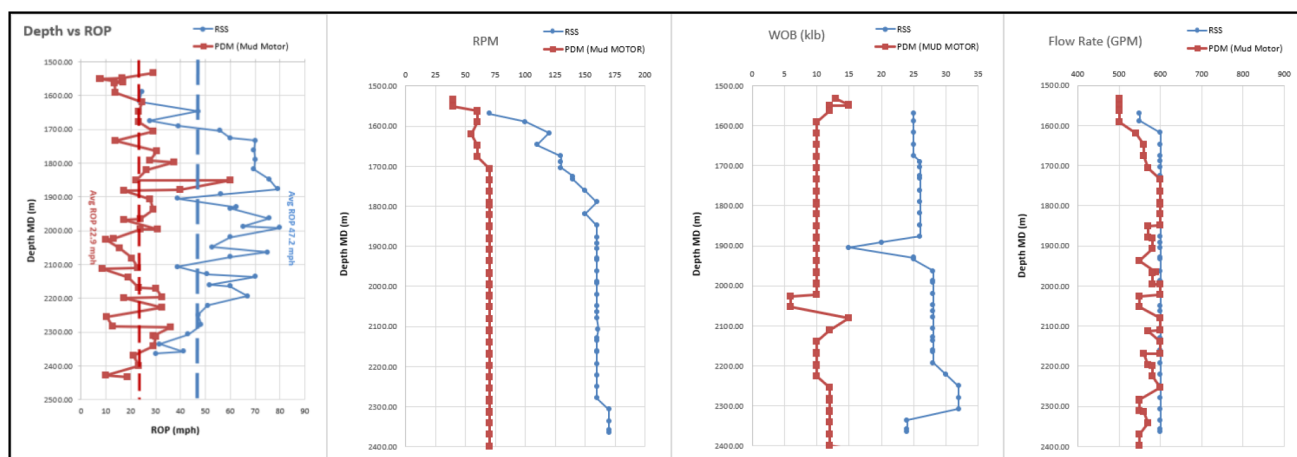


Figure 7—Performance Comparison (ROP, RPM, WOB, GPM) between Target Well-2 and Offset Well-1

The ROP comparison graph reveals that RSS push-the-bit consistently outperforms the PDM system in this S-Type Well. RSS push-the-bit achieved a maximum ROP of 79 m/hr, minimum of 24 m/hr, and an average of 47.2 m/hr, and the PDM motor demonstrated significantly lower performance, with a maximum ROP of 60 m/hr, minimum of 7.5 m/hr, and an average of only 22.9 m/hr. In terms of RPM, RSS push-the-bit operated within a range of 70 to 170 RPM, while the PDM motor was limited to 40 to 70 RPM. This higher rotational speed not only helps enhance rate of penetration but also helps improve hole cleaning and toolface control while rotating with RSS push-the-bit. The PDM motor need to slide with stop rotating the string for control the tool face direction which is critical in directional drilling. For WOB, the PDM required more variable range (10 to 23 klb), indicating higher force was necessary to maintain performance and tool face direction. In contrast, RSS push-the-bit operated effectively with a more consistent WOB between 25 and 32 klb, reflecting its ability to sustain performance with a more controlled load, time and efficiency. Regarding flow rate, both systems operated within a similar range. RSS push-the-bit maintained flow between 550–600 GPM, while the PDM ranged from 500–600 GPM. This indicates that although hydraulic support was similar, RSS push-the-bit helped deliver significantly improved performance with the same fluid energy input, RSS push-the-bit gives more stability Flow rate range which is effect to Hole cleaning and borehole quality

In summary, the visual and data-driven comparison clearly demonstrates that RSS push-the-bit offers advantages in directional drilling performance. The RSS push-the-bit drilled faster, more controlled drilling, and more effective compared to conventional mud motors.

Comparison Against Several Offset Wells.

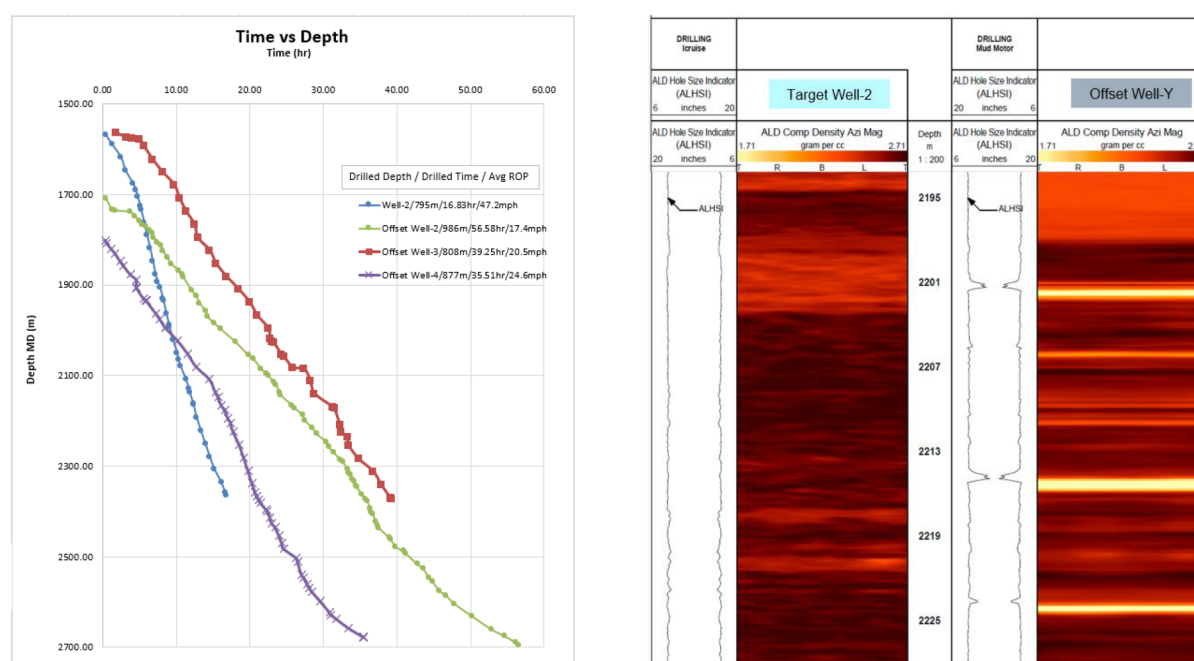


Figure 8—(Left) Performance Comparison (Time vs Depth) against several offset wells; (Right) Caliper and Density Image displays Borehole Quality

Automation Drilling Platform

Activating automation drilling platform into the deployed intelligent RSS push-the-bit had enabled a lot of benefits, primarily in providing better and accurate control towards drilling parameters, which allows directional drillers to focus on design improvements and less of doing repetitive and manual tasks. The whole drilling automation algorithm put focus on achieving as close as possible placement of actual trajectory against well-plan input, with steady and consistent result of DLS. There are classifications on the level of

automation drilling platform that can be applied, it is segmented as how much human interferences are involved. The basic level will only provide recommendations in the format of notifications that need to be acknowledged by directional driller; while the sophisticated level will do full automation loop, in which the recommendations given are directly transmitted to intelligent RSS downhole and will not accept any human interference. In this case studies, the middle level of automation is applied, the algorithm of recommendation will always be accepted, unless there is a significant deviation in which directional driller can take override and adjust the algorithm's input from time to time.

Referring to drilling-automation's performance tracker in both subject Wells (Fig.9), automated cruise trajectory had been seen successfully active for ~83% of the entire drilling stand of the wells. The rest of them were tracked as manual downlink or human input for steering command. This translates to an average of ~638 m.MD drilling interval was steered fully automated without human interference. Table on the right summarized the downlink or automated steering command given from the digital platform.

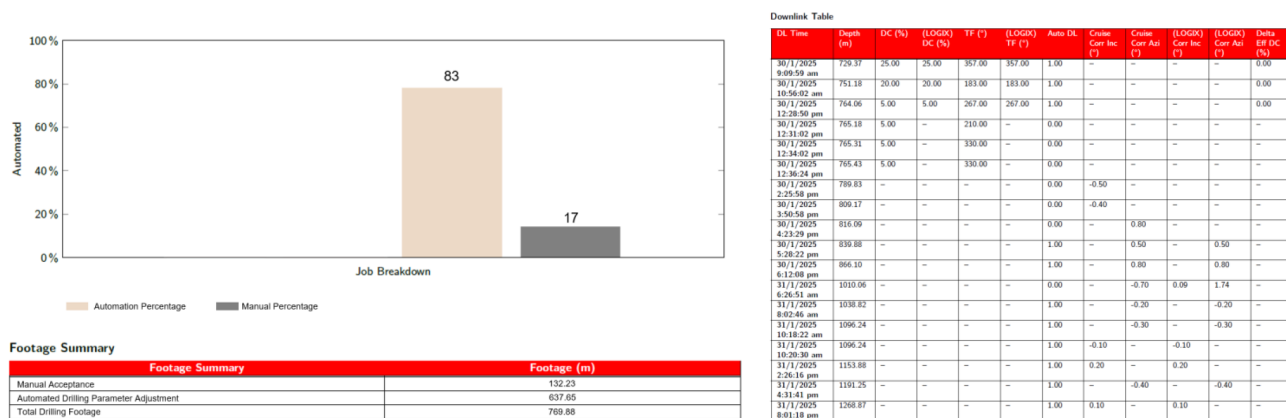


Figure 9—(Left) Drilling Automation Performance Tracker; (Right) Sample List of Drilling Parameter Adjustments by Automation Algorithm

Final result can be seen in (Fig.10) as the automation algorithm provided the corrected inclination and azimuth by adjusting toolface and duty cycle (RSS push-the-pit effective pad pushing bias). In overall, corrected graph has shown that drilling automation has able to maintain stability in achieving and maintaining drilling output against original well plan. Automation in both wells had yielded the result of center-to-center less than 1 meter, with extremely low DLS of less than 0.5 deg/30meter, meaning very less tortuosity had successfully achieved. In the next stage of operations, running casing and completion went smoothly benefiting from good borehole quality drilled with automation.

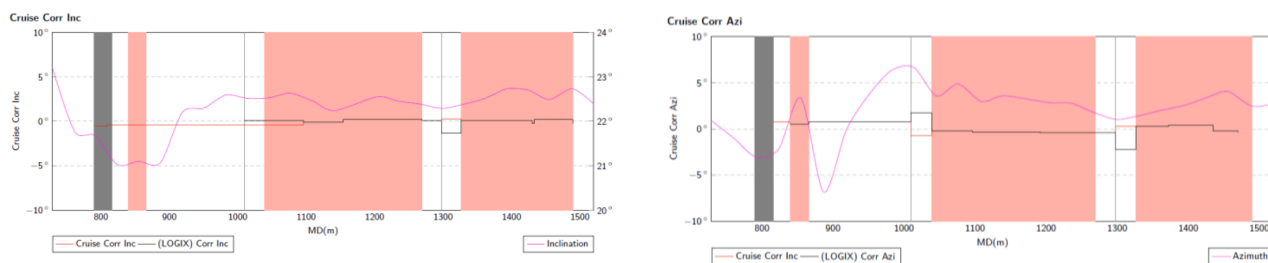


Figure 10—Result of Corrected Inclination (Left) and Corrected Azimuth (Right) from Drilling Automation; plotted against Well Plan

Conclusion

This case studies have elaborated the success run of deploying relatively new technology of RSS push-the-bit and an automation platform in a factory wells onshore project. Proper engineering review and

simulation in the pre-job planning of RSS push-the-bit had highlighted the importance of mitigating all risks (specifically where this subject field had high risk of total loss). RSS push-the-bit had enabled the significant improvement of drilling activities in both wells, Target Well-1 achieved ~65 m/hr ROP and Target Well-2 achieved ~80 m/hr. This had exceeded 40% of more ROP compared to the common execution with PDM. Automation had aid precise and accurate placement of < 0.5 meter center-to-center against original well plan, saving total of ~2 days in operations rig days which are equivalent to ~130K USD of onshore operational cost. In overall, this deployment of novel method in factory wells onshore drilling operations has proven to be a fit-for-purpose solution and become a game changer in speeding up well to well deliverables, making mature drilling project to be economically feasible.

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References

- Adiandra, K., Christyananta, T., Manurung, V., Hutabarat, D., Munoz, D., Greenwood, J., Ridhanta, I.N.G., Kurniawan, A., Sany, M.F., Putra, A.M., "Autonomous Steering Digital Platform and Push-the-Bit Rotary Steerable System Achieves Significantly Increased Hole Quality and Time Savings, A Case Study from the First Run in Offshore North-West-Java Indonesia." Paper presented at the Offshore Technology Conference, Rio De Janeiro, Brazil, October 2025.
- Aprilian, S., and Maria R., "Improving Production Monitoring and Surveillance by Using an Integrated Production Data Management." Paper presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition, Jakarta, Indonesia, September 2003. doi: <https://doi.org/10.2118/80523-MS>
- Fansuri, T. Miftah, A., Parsaulian, S., Giyatno, Riviana R., Yanuardi, A., Mesia P., Latissa, and Suherman, A.T., "The Benefits of Real Time Monitoring Application Artificial Lifting Parameter for Surveillance, Optimization, and Cost Reduction in Mature Field, Case Study Prabumulih Field." Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, November 2021. doi: <https://doi.org/10.2118/207596-MS>
- Galih, N.P., Rudhy, S., Asep, S., Cahya, N.M., and Humam, H.A., "A Novel Method: Injection Dynamic - Transient Acoustic Pressure Thermal Logging to Increase the Effectiveness of Water Shutoff and Remediation Operation in Depleted Production Wells." Paper presented at the ADIPEC, Abu Dhabi, UAE, October 2023. doi: <https://doi.org/10.2118/216558-MS>